

LABTASK 1:

```
//QUESTION 1
class Distance{
    private:
    int feet;
    int inch;

    public:
    Distance (){
        feet =0;
        inch =0;
    }
    Distance operator + (Distance &d){
        Distance temp;
        int inches=inch + d.inch;
        temp.feet=feet + d.feet + inches/12;
        temp.inch =inches % 12;
        return temp;
    }

    bool operator == (Distance &d){
        int total1 = this->feet * 12 + this->inch;
        int total2 = d.feet * 12 + d.inch;

        return total1==total2;
    }
}
```

```

void setData(int f, int i){
    |   |   feet = f;
    |   |   inch = i;
}
void showData(){
    |   cout<<feet<<" feet "<<endl;
    |   cout<<inch<<" inches "<<endl;
}
};

```

```

};
int main(){
    Distance d1,d2,d3,d4;
    d1.setData(2,5);
    d2.setData(6,10);
    d4.setData(2,5);

    cout<<"DISTANCE 1: "<<endl;
    d1.showData();
    cout<<"DISTANCE 2: "<<endl;
    d2.showData();
    d3 = d1+d2;
    cout<<"D1 + D2: "<<endl;
    d3.showData();
    if(d1==d4){
    |   |   cout<<"D3 AND D4 ARE EQUAL"<<endl;
    }
    else{
    |   cout<<"D3 AND D4 ARENOT EQUAL"<<endl;
    }
    |   return 0;
    }
}

```

```
PS C:\Sem 2\OOP LABS\lab10>
DISTANCE 1:
2 feet 5 inches
DISTANCE 2:
6 feet 10 inches
D1 + D2:
9 feet 3 inches
D3 AND D4 ARE EQUAL
PS C:\Sem 2\OOP LABS\lab10>
```

LABTASK 2:

```
// QUESTION 2
class student{
private:
    char* name;
    int roll_no;
public:
    student(const char *n, int r){
        name = new char[strlen(n)+1];
        strcpy(name, n);
        roll_no = r;
    }
    ~student(){
        delete [] name;
    }

    void showData(){
        cout<<"NAME: "<<name<<endl;
        cout<<"ROLL NUM: "<<roll_no<<endl;
    }
};
```

```
int main(){
    student obj_1("Usman", 101);
    student obj_2=obj_1;

    obj_1.showData();
    obj_2.showData();
    return 0;
}
```

NAME: Usman

ROLL NUM: 101

NAME: Usman

ROLL NUM: 101

PS C:\Sem 2\OOP LABS\lab10> █

LABTASK 3:

```
//QUESTION 3
class student{
    private:
        char* name;
        int roll_no;
    public:
        student(const char *n, int r){
            name = new char[strlen(n)+1];
            strcpy(name, n);
            roll_no = r;
        }
        ~student(){
            delete [] name;
        }

        student(student &obj){
            name = new char[strlen(obj.name)+1];
            strcpy(name, obj.name);
            roll_no = obj.roll_no;
        }
        void showData(){
            cout<<"NAME: "<<name<<endl;
            cout<<"ROLL NUM: "<<roll_no<<endl;
        }
};
```

```
int main(){
    student obj_1("Usman", 101);
    student obj_2(obj_1);

    obj_1.showData();
    obj_2.showData();
    return 0;
}
```

PS C:\Sem 2\OOP LABS\lab10> g++

NAME: Usman

ROLL NUM: 101

NAME: Usman

ROLL NUM: 101

PS C:\Sem 2\OOP LABS\lab10> |